Jonas Elias El Gammal

Cosmologist and scientific machine-learning researcher

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PROFILE

Postdoctoral researcher working at the intersection of cosmology, gravitational-wave astronomy, Bayesian inference, and scientific machine learning. Developer of open-source tools for accelerating expensive likelihood-based inference and modelling gravitational-wave signals.

RESEARCH EXPERIENCE

National Institute of Chemical Physics and Biophysics (KBFI/NICPB) - Postdoctoral Researcher | 2026-present
Tallinn, Estonia

University of Insubria - Postdoctoral Researcher | 2025-2026

- Accelerated Pulsar Timing Array inference using GPUs and differentiable programming with JAX as part of the EPTA collaboration.

CERN Theory Department - Visiting Researcher | 2024

- Developed a LISA data-analysis pipeline for second-order scalar-induced gravitational waves from inflation with M. Pieroni and G. Franciolini.

EDUCATION

University of Stavanger - PhD in Physics | 2021-2025
Thesis: *Cutting edge data analysis for gravitational wave detection with LISA*.

Grenoble INP, PHELMA - Erasmus Exchange | 2019-2020
Specialisation in nuclear engineering.

RWTH Aachen University - MSc Physics | 2018-2020
Grade 1.3. Thesis: *Accelerating Bayesian Inference of Expensive Likelihoods with Gaussian Processes*.

RWTH Aachen University - BSc Physics | 2015-2018
Grade 1.7. Focus on astroparticle physics and cosmology.

SELECTED PUBLICATIONS

Are PTA measurements sensitive to gravitational wave non-Gaussianities? (2026)
C. Cecchini, J. El Gammal, G. Franciolini, M. Pieroni

Accelerating LISA inference with Gaussian processes (2025)
J. El Gammal, R. Buscicchio, J. Torrado, G. Nardini

Reconstructing Primordial Curvature Perturbations via Scalar-Induced Gravitational Waves with LISA (2025)
LISA Cosmology Working Group

Parallelized Acquisition for Active Learning using Monte Carlo Sampling (2023)
J. Torrado, N. Schöneberg, J. El Gammal

Fast and Robust Bayesian Inference using Gaussian Processes with GPry (2022)
J. El Gammal et al.

OPEN-SOURCE SOFTWARE

GPry - Bayesian inference for expensive likelihoods using Gaussian-process surrogate modelling and active learning.

SIGWAY - Simulation and inference tools for second-order scalar-induced gravitational waves from inflation.

discoverysamplers - Unified interfaces connecting Discovery PTA models to multiple sampling backends.

fastropop - JAX-enabled simulation of supermassive black-hole-binary populations and nanohertz gravitational-wave signatures.

TEACHING

Numerical Methods, University of Stavanger - full bachelor-level course (2023, 2024). Python programming; numerical differentiation, integration, ODEs, and PDEs.

Electromagnetism and Special Relativity, University of Stavanger - full bachelor-level course (2023).

Physics and data-processing tutorials, RWTH Aachen University (2017-2019).

SKILLS

Research: Bayesian inference, Gaussian processes, uncertainty quantification, active learning, time-series analysis, cosmology, and gravitational-wave data analysis.

Software: Python, JAX, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow, SQL, C++, Git, Docker, Linux, SLURM, HPC, CI/CD, and scientific package development.

Languages: German (native), English (near-native), French (B2/C1).